

Experiment 1

Welcome to the world of Programming!

In this challenge, you will learn how to print "Hello World!" in standard output (stdout).

Output Format

You must print two lines of output:

Print "Hello, World." on the first line (without quotes).

Print "Hello, Java." on the second line (without quotes).

Sample Output

Hello, World.

Hello, Java.

Experiment 2

In this task, you have to print n stars * vertically and n horizontally.

Input Format

The first line of input contains n.

Output Format

print the stars in horizontal and vertical direction.

Example 1

Input

3

Output

* * *

*

*

*

Experiment 3

In this task, you need to print a right-angled triangle of height 3 (lines) and base 5 (stars).

```
*  
***  
*****
```

Input Format

No input.

Output Format

Print the triangle of height 3 and base 5 as shown.

Example 1

Input

No input

Output

```
*  
***  
*****
```

Experiment 4

Given two number a and b, find their HCF.

What Is HCF?

HCF or Highest Common Factor is the greatest common divisor between two numbers.

Input Format

The first line of input contains the integers a and b.

Output Format

The output should be the hcf of a and b.

Example 1

Input

75 90

Output:

15

Explanation

$75 = 3 * 5 * 5$ and $90 = 2 * 3 * 3 * 5$. So the common factors are $3*5 = 15$.

Example 2

Input

25 50

Output

25

Experiment 5

Write a program that prompts the user to input an integer and then outputs the number with the digits reversed.

For example, if the input is 12345, the output should be 54321.

Note: Input number will not have any trailing zeros.

Input Format

The first line of input contains the integer n.

Output Format

The output should be the reverse of n.

Example 1

Input

12345

Output:

54321

Experiment 6

For an integer N, your task is to calculate the sum of first N natural numbers.

Input User Task: You have to create a function NumberSum() which takes the integer (large value possible) N as a parameter and returns the sum of first N natural numbers..

Input Format

First line is an integer N

Output Format

Print the sum of the first N natural numbers.

Example 1

Input

5

Output

15

Experiment 8

Write a program to find the factorial of a number. Create a function factorial() to do the operation.

Experiment 9

Write a program to find the sine and cosine of a given angle in degrees. Create two methods sine() and cosine().

Experiment 10

PEAK ELEMENT

An element is called a peak element if its value is not smaller than the value of its adjacent elements(if they exists).

For first and last element consider only the element adjacent to it.

Given an array arr of size n, find the index of first peak element.

If peak element does not exist, print -1.

Input Format

First line contains an integer n.

Second line contains n space separated integers.

Output Format

Return the index of the peak element in zero based indexing if present, else return -1

Example 1

Input

4

5 10 20 15

Output

2

Explanation

20 is greater than both of its neighbours. Hence 20 is the peak element, So output is 2

Example 2

Input

7

10 20 15 2 23 64 67

Output

1

Explanation

20 is greater than both of its neighbors. Hence 20 is the peak element, So output is 1

Experiment 11

You are given an integer array `nums` of size `n` where the largest integer is unique. Your task is to determine whether the largest element in the array is at least twice as much as every other number in the array. If it is, print the index of the largest element, or print -1 otherwise.

Input Format

The first line of the input contains the number `n`(length of the array),

The next line contains `n` space separated integers denoting the elements of the array.

Output Format

Print the index of the element if it satisfies the condition else print -1

Example 1

Input

4

3 6 1 0

Output

1

Explanation

Here, 6 is the largest integer, and for every other number in the array `x`, 6 is more than twice as big as `x`. The index of value 6 is 1, so we return 1

Example 2

Input

4

1 2 3 4

Output

-1

Explanation

4 isn't at least as big as twice the value of 3, so we return -1.

Experiment 12

Given an unsorted array of size N find the 2nd largest element from the array without sorting the array.

Input Format

The first line contains a single integer N.

The second line consists of N integers of the array.

Output Format

Print the second largest number in the new line.

Example 1

Input

6

3 2 1 5 6 4

Output

5

Explanation

5 is the second largest element of the array.

Example 2

Input

9

10 1 7 2 5 6 8 9 3

Output

9

Explanation

9 is the Second largest element in the array

Experiment 13

Given an array of integers, arr, and a positive integer k, determine the number of (i,j) pairs in the array where $i < j$ and $\text{arr}[i] + \text{arr}[j]$ is divisible by k.

Input Format

Input consists of two lines.

The first line contains 2 space-separated integers, n (size of array) and k.

The second line contains n space-separated integers, each a value of $\text{arr}[i]$.

Output Format

Print the number of pairs $a[i] + a[j]$ divisible by k, where $i < j$.

Example 1

Input

6 3

1 3 2 6 1 2

Output

5

Explanation

Here are the 5 valid pairs when $k = 3$:

$$(0,2) = \text{arr}[0] + \text{arr}[2] = 1 + 2 = 3$$

$$(0,5) = \text{arr}[0] + \text{arr}[5] = 1 + 2 = 3$$

$$(1,3) = \text{arr}[1] + \text{arr}[3] = 3 + 6 = 9$$

$$(2,4) = \text{arr}[2] + \text{arr}[4] = 2 + 1 = 3$$

$$(4,5) = \text{arr}[4] + \text{arr}[5] = 1 + 2 = 3$$

Example 2

Input

4 5

1 3 2 6

Output

1

Explanation

Here is the 1 valid pairs when $k = 5$:

$$(1,2) = \text{arr}[1] + \text{arr}[2] = 2 + 2 = 5$$

Experiment 14

Given an array Arr of N elements and a integer K. Your task is to print the position of first occurrence of K in the given array. If the number is not found, print -1.

Note: Position of first element is considered as 1.

Input Format

The first line contains two integers N and K.

The second line contains N spaced integers, elements of Arr.

Output Format

Print the position of first occurrence of K.

If K is not present in the array, print -1.

Example 1

Input

5 16

9 7 2 16 4

Output

4

Explanation

K = 16 is found in the given array at position 4.

Example 2

Input

6 6

5 7 7 8 8 10

Output

-1

Explanation

6 doesn't occur in the given array, hence we print -1.

Experiment 15

Given an integer array `nums` and an integer `k`, print the number of non-empty subarrays that have a sum divisible by `k`.

A subarray is a contiguous part of an array.

Input Format

Input consists of two lines.

The first line contains two integers `n` (size of the array) and `k`.

The second line contains `n` spaced integers.

Output Format

Print count of subarrays divisible by `k`

Example 1

Input

6 5

4 5 0 -2 -3 1

Output

7

Explanation

There are 7 subarrays with a sum divisible by `k = 5`:

[4, 5, 0, -2, -3, 1]

[5]

[5, 0]

[5, 0, -2, -3]

[0]

[0, -2, -3]

[-2, -3]

Example 2

Input

4 2

4 5 0 -2

Output

4

Explanation

There are 4 subarrays with a sum divisible by $k = 2$:

[4]

[0]

[0 -2]

[-2]

Experiment 16

Given an array of n integers where $n > 1$, return an array output such that $\text{output}[i]$ is equal to the sum of all the elements of nums except $\text{nums}[i]$. All integers in array are greater than 0.

Input Format

First line consists of an integer n which is the number of elements in array

Next n lines correspond to n elements of array

Output Format

Return the resultant array as asked in question.

Example 1

Input

4

4 3 2 10

Output

15 16 17 9

Explanation

After updation,

$a[0]=15$ i.e., $3+2+10$

$a[1]=16$ i.e., $4+2+10$

$a[2]=17$ i.e., $4+3+10$

$a[3]=9$ i.e., $4+3+2$

Experiment 17

Given an array h representing heights of buildings. You have to count the buildings which will see the sunrise (Assume : Sun rise on the side of array starting point).

Input Format

line 1: contains an integer n denoting size of array.

line 2: contains n spaced integers denoting elements of array.

Output Format

Print a single integer denoting the number of buildings which will see the sunrise.

Example 1

Input

5

7 4 8 2 9

Output

3

Explanation

7 can watch the sunrise, 4 can't, then 8 can but 2 can't and lastly 9 can watch the sunrise too, hence 3 buildings can watch the sunrise.

Example 2

Input

6

6 2 8 4 11 13

Output

4

Explanation

Only buildings of height 6, 8, 11 and 13 can see the sun, hence output is 4.

Example 3

Input

4

1 1 1 1

Output

1

Explanation

Only 1st building will be able to see.

Experiment 18

Marc loves cupcakes, but he also likes to stay fit. Each cupcake has a calorie count, and Marc can walk a distance to burn those calories.

On eating the j^{th} cupcake with c calories, he must walk at least $(2^j) \cdot c$ miles to maintain his weight.

For example, let $\text{calorie}=[5,10,7]$

If he eats the cupcakes in the order shown, the miles he will need to walk are $(2^0 \cdot 5) + (2^1 \cdot 10) + (2^2 \cdot 7) = 5 + 20 + 28 = 53$. This is not the minimum, though, so we need to test other orders of consumption.

In this case, our minimum miles is calculated as $(2^0 \cdot 10) + (2^1 \cdot 7) + (2^2 \cdot 5) = 10 + 14 + 20 = 44$.

Given the individual calorie counts for each of the cupcakes, determine the minimum number of miles Marc must walk to maintain his weight.

Note that he can eat the cupcakes in any order.

Input Format

The first line contains an integer n , the number of cupcakes.

The next line inputs the calories of n cupcakes.

Output Format

Print the minimum miles necessary.

Example1

Input

3

1 3 2

Output

11

Explanation

Let us say the number of miles Marc must walk to maintain his weight is miles.

He can minimize miles by eating the $n=3$ cupcakes in the following order:

Eat the cupcake with $c_2=3$ calories, so $\text{miles} = 0 + (3 \cdot 2^0) = 3$.

Eat the cupcake with $c_1=2$ calories, so $\text{miles}=3+(2 \cdot 2^1)=7$.

Eat the cupcake with $c_0=1$ calories, so $\text{miles}=7+(1 \cdot 2^2)=11$.

We then print the final value of miles, which is 11, as our answer.